



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
REVIEW TEST
15-05-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS)

Time : 3 Hours

PAPER-1

Maximum Marks : 180

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- | | |
|------------------------------------|--|
| • Boltzmann constant | $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$ |
| • Coulomb's law constant | $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$ |
| • Universal gravitational constant | $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ |
| • Speed of light in vacuum | $c = 3 \times 10^8 \text{ ms}^{-1}$ |
| • Stefan-Boltzmann constant | $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$ |
| • Wien's displacement law constant | $b = 2.89 \times 10^{-3} \text{ m-K}$ |
| • Permeability of vacuum | $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$ |
| • Permittivity of vacuum | $\epsilon_0 = \frac{1}{\mu_0 c^2}$ |
| • Planck constant | $h = 6.63 \times 10^{-34} \text{ J-s}$ |

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen and both of which are correct.

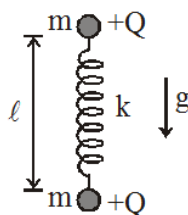
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

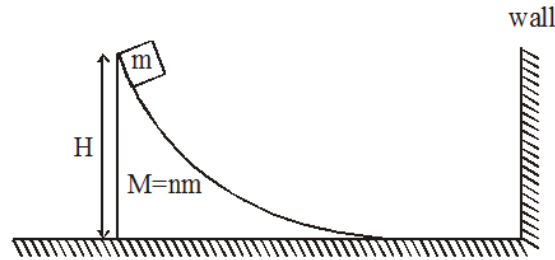
1. Two identical balls each of mass m carries equal charge Q are connected to a massless spring of spring constant k and released from rest. Initially spring was in its natural length ℓ . Take $k = \frac{2Q^2}{3\pi\epsilon_0\ell^3}$, then select correct option(s)



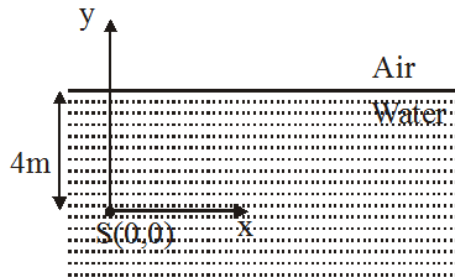
- (A) Maximum distance between balls in motion will be $\frac{3\ell}{2}$.
- (B) Both balls will perform oscillation in frame of centre of mass but these oscillations will not be simple harmonic.
- (C) Acceleration of centre of mass of system will be decreasing with time.
- (D) Acceleration of centre of mass of system will be increasing with time.

Space for Rough Work

2. All the surface are smooth and collisions are elastic in nature. The system is released from rest. Then



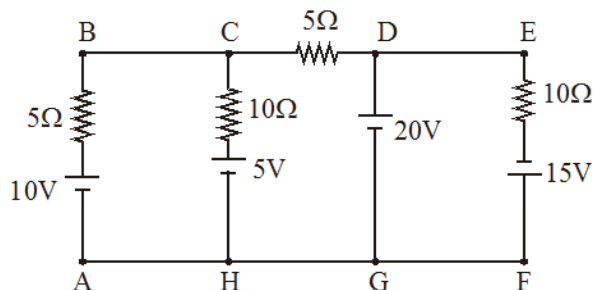
- (A) When block first time slide down wedge and reaches horizontal surface velocity of centre of mass of block-wedge system have zero velocity.
- (B) After a long time for $n < 3$ velocity of centre of mass of block wedge system will be $\sqrt{\frac{8ghn}{(n+1)^3}}$.
- (C) After a long time for $4 > n > 3$ velocity of centre of mass of block wedge system will be $\sqrt{\frac{32ghn(n-1)^2}{(n+1)^5}}$.
- (D) Velocity of centre of mass of wedge system will not have vertical component.
3. A point source S is placed in water at origin (0, 0) and an observer O is in air at (7m, 7m) as shown in figure. (Given : $\mu_{\text{Water}} = 4/3$)



- (A) Ray starting from source along $3\hat{i} + 4\hat{j}$ vector will reach observer after refraction.
- (B) Image of object seen by observer will be at line $4y - 3x = 7$.
- (C) Co-ordinate of image seen by observer is $\left(\frac{21}{16}\text{m}, \frac{175}{64}\text{m}\right)$.
- (D) Ray starting from source along $4\hat{i} + 3\hat{j}$ vector will reach observer after refraction.

Space for Rough Work

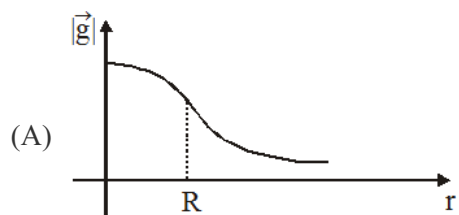
4. In given circuit all batteries are ideal than:



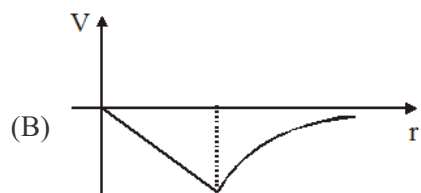
- (A) Current in branch CD is 1.4 A from D to C.
 (B) 10V battery is charging, consuming 6W power.
 (C) Rate of heat production in branch CH is 8.4 J/s.
 (D) In this circuit only 20V battery is discharging supplying power of 98W.

Space for Rough Work

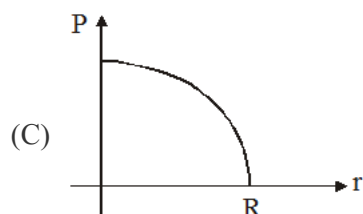
5. Assuming the earth to be uniform sphere of radius R and mass M and neglecting rotational motion select correct option(s)



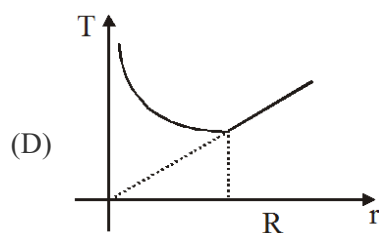
$|\vec{g}|$: Acceleration due to gravity, r : distance from centre



v : gravitational potential



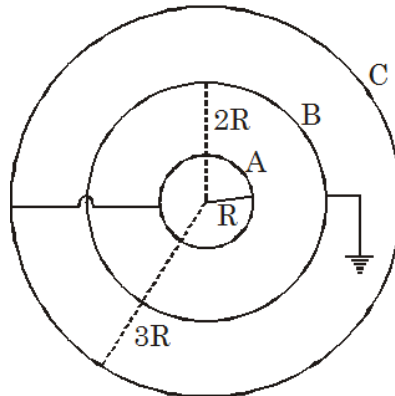
P : gravitational pressure.



T : time period of simple pendulum.

Space for Rough Work

6. Three conducting spherical shell A,B and C has radius R , $2R$ and $3R$ respectively. Conductor B is connected to earth and A and C are connected by conducting wire. If charge Q is given to conductor C then :-



- (A) Charge on outer surface of conductor A is $\frac{2Q}{11}$.
- (B) Charge on inner surface of conductor C is $\frac{6Q}{11}$.
- (C) Charge on conductor B is $-\frac{8Q}{11}$.
- (D) Charge on outer surface of conductor C is $\frac{3Q}{11}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

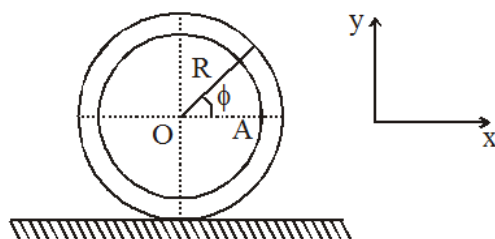
Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

A Non-conducting ring of mass m and radius R is placed on a rough horizontal surface as shown in figure.



It has linear charge density $\lambda = \lambda_0 \sin \phi$ where λ_0 is constant and angle ϕ is angle of radius vector with line OA.

7. Electric field intensity at centre of ring is :

(A) $\frac{\lambda_0}{8\pi\epsilon_0 R} (\hat{i})$ (B) $\frac{\lambda_0}{4\epsilon_0 R} (-\hat{j})$ (C) $\frac{\lambda_0}{8\pi\epsilon_0 R} (-\hat{j})$ (D) $\frac{\lambda_0}{4\pi^2\epsilon_0 R} (+\hat{j})$

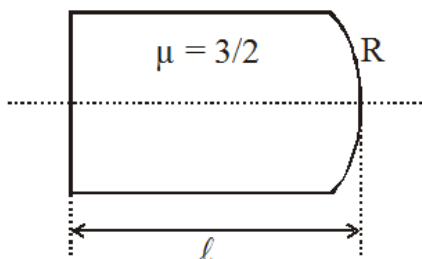
8. When external uniform electric field $E = E_0 \hat{i}$ is switch on at $t = 0$ then speed of ring after it rotates by angel $\pi/2$ will be (assuming friction is sufficient to prevent slipping)

(A) $R\sqrt{\frac{3\pi\lambda_0 E_0}{m}}$ (B) $R\sqrt{\frac{\lambda_0 E_0}{m}}$ (C) $R\sqrt{\frac{\pi\lambda_0 E_0}{m}}$ (D) $\pi R\sqrt{\frac{2\lambda_0 E_0}{m}}$

Space for Rough Work

Paragraph for Questions 9 and 10

A thick plano-convex lens is made of material having refractive index $\mu = \frac{3}{2}$. When its plane surface is silvered and an object is placed 60 cm away from its convex surface then final image forms at object itself. While when its convex surface is silvered and object is placed at distance $\frac{20}{3}$ cm away from plane surface then final image again forms at object itself.



9. Radius of curvature of convex surface is :
- (A) 10 cm (B) 15 cm (C) 25 cm (D) 30 cm
10. Thickness of lens is
- (A) 10 cm (B) 15 cm (C) 20 cm (D) 25 cm

Space for Rough Work

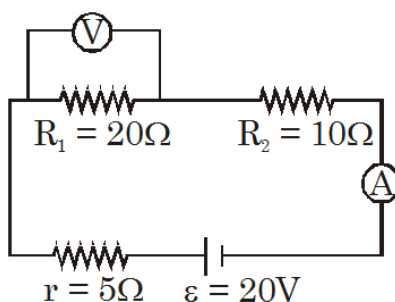
SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

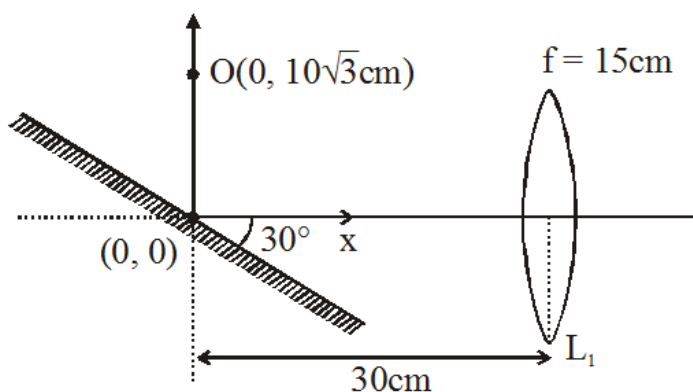
1. In given circuit reading of ammeter and voltmeter are $\frac{2}{3}$ A and $\frac{20}{3}$ V respectively. If resistance of voltmeter is R_V and resistance of ammeter is R_A then find $R_V - R_A$ (in Ω).



2. A uniform cloud of dust particle approach a planet from a great distance with velocity v_0 relative to the planet. Radius of planet is R and escape velocity from surface of the planet is $v_e = 2v_0$. Length of the cloud along direction of motion is $\ell = R$, Cross-section of cloud perpendicular to it's direction of motion is very large and it's density is $\frac{4}{3000}$ times density of planet. If mass of planet m is very large compared to the mass of cloud then mass of the dust particle that planet will collect during passage through the cloud is $\frac{m}{10x}$. The value of x is :

Space for Rough Work

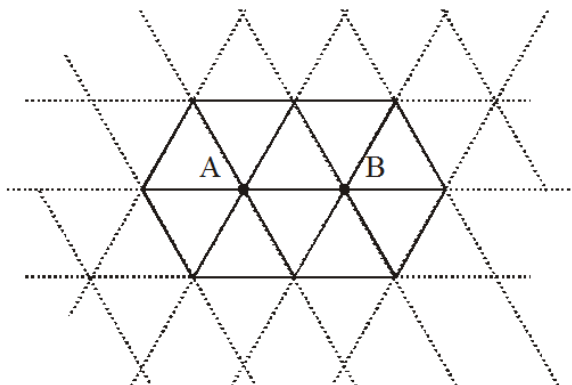
3. A plane mirror M_1 is placed at angle 30° with x-axis and a lens L_1 of focal length 15 cm is placed 30 cm away from it as shown in figure. A source O is placed at $[0, 10\sqrt{3}\text{cm}]$. If co-ordinate of image formed after reflection from plane mirror then refraction from lens is (x,y) then value of $\sqrt{3}\frac{x}{y}$ is.



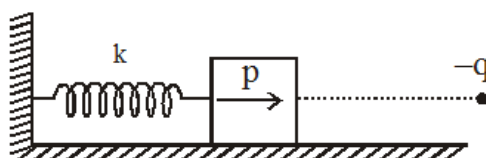
4. A square non-conducting sheet of side a has uniform charge density σ . A point charge Q is placed at perpendicular distance $d = a$ above one of it's edges. Then component of electric force acting on sheet perpendicular to it's plane is F . Find $\frac{Q\sigma}{F\epsilon_0}$.

Space for Rough Work

5. In an infinite grid of triangular cell, resistance of each branch is $R = 210\Omega$. The equivalent resistance of circuit between the terminal A and B is $x\Omega$. Find x.

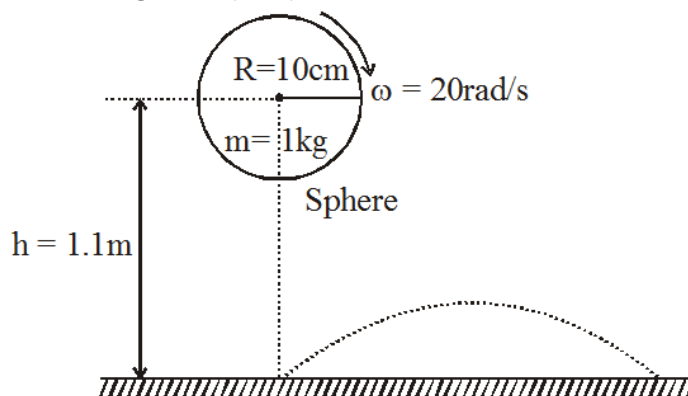


6. A small block having electric dipole moment p is placed on a frictionless horizontal table is attached to one end of a spring of force constant k as shown. Now a point charge $-q$ is brought very slowly from a great distance towards block along line coinciding the axis of the spring. Extension in spring when block stick to charge is $\left[\frac{apq}{b\pi\epsilon_0 k} \right]^{\frac{1}{4}}$ where a and b are smallest possible integer than $\frac{a-b}{100}$ is.

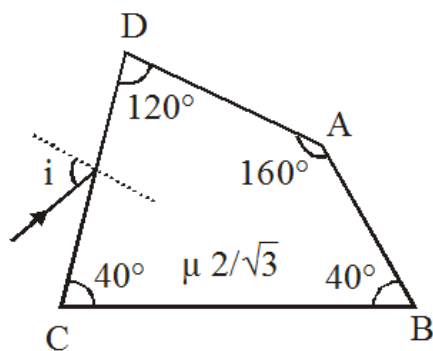


Space for Rough Work

7. A ball mass $m = 1\text{kg}$ and radius $R = 10\text{cm}$ is rotating with angular velocity $\omega = 20\text{ rad/s}$ about horizontal axis. It is dropped from a height $h = 1.1\text{m}$ on a rough horizontal surface ($\mu_k = 0.5$) as shown in figure. If coefficient of restitution for collision is $e = 0.5$ then find horizontal distance travelled by ball between points of 1st and 2nd collision with ground (in m)



8. A monochromatic light ray incident on block ABCD having R.I $\mu = \frac{2}{\sqrt{3}}$ at angle of incidence $i = \sin^{-1}\left[\frac{1}{\sqrt{3}}\right]$ as shown in figure. Find angle between incident and emergent ray (in degree)



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

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- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Which of the following is(are) correct ?

- (A) Number of geometrical isomers of ionisation isomers of $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{Cl}_2]\text{Br}$ is 5 (except the given compound)
- (B) In $\text{Fe}_2(\text{CO})_9$, $d_{\text{C-O}}$ (bridging) < $d_{\text{C-O}}$ (terminal)
- (C) IUPAC name of $[(\text{Cl}_3\text{Sn})_2\text{RhCl}_2\text{Rh}(\text{SnCl}_3)_2]^{4+}$ is Di- μ -chloridotetrakis(trichlorostannito)dirhodium(I) ion
- (D) Complexes $[\text{Co}(\text{NO}_2)_6]^{4+}$ and $[\text{CoF}_6]^{3-}$ are inner orbital complex and outer orbital complex respectively

Space for Rough Work

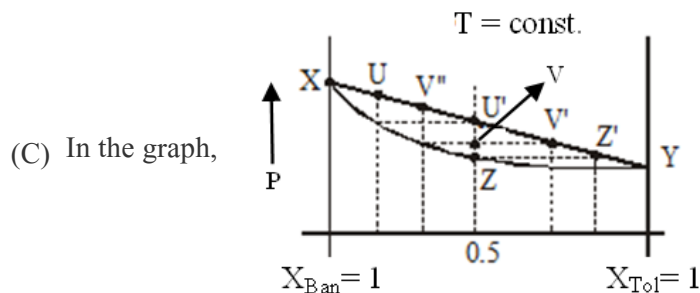
2. Which statement(s) can be possible for homonuclear diatomic neutral molecules formed by any 2nd period element.
- (A) Atleast one paramagnetic species has only σ bonds
- (B) Atleast one diamagnetic species has only σ bonds
- (C) Atleast one diamagnetic species has only π bonds
- (D) Atleast one paramagnetic species has only π bonds
3. 250 ml of 1M I_2 is separately reacted with 0.5M Cu_2S solution, 0.5M CuS solution and 0.5 M $S_2O_3^{2-}$ solution causing production of Cu^{+2} and SO_4^{2-} in the first two and $S_4O_6^{2-}$ in the last case along with iodide ions. Which of the following option is/are correct assuming 100% completion of the reaction. (I = 127)
- (A) 100 ml of Cu_2S solution will be consumed.
- (B) 125 ml of CuS solution will be consumed.
- (C) 500 ml of $S_2O_3^{2-}$ solution will be consumed.
- (D) Equivalent weight of I_2 in each of the reaction will be 127.

Space for Rough Work

4. 1 mole each of benzene and toluene are mixed (ideal solution) at a given temperature. A total of 1 mole has vaporised, at given pressure. Then, which of the following must be true for this equilibrium mixture?
[Given : $P_{\text{Benzene}}^0 = x$; $P_{\text{Toluene}}^0 = y$]. Assume X, U, V'', U', Z, V', Z' are various points on the curve.

(A) Total vapour pressure $< \left[\frac{x+y}{2} \right]$

(B) Total vapour pressure $= \sqrt{xy}$

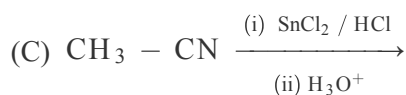
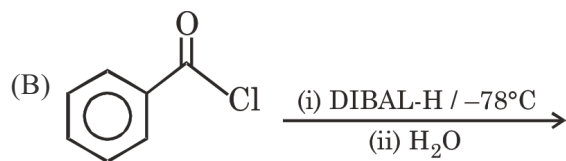
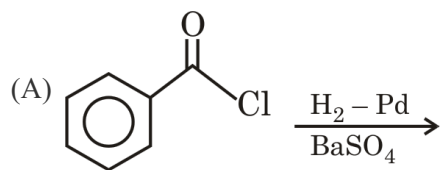


the vapour pressure of equilibrium mixture may correspond to point V''.

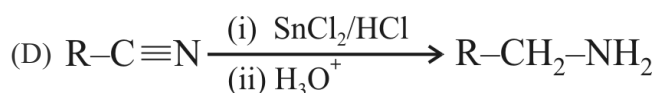
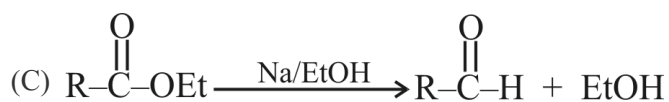
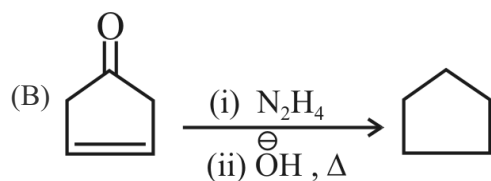
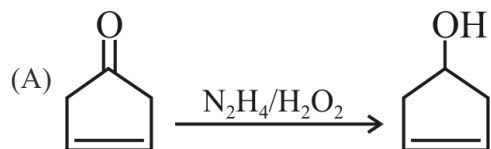
- (D) If in the graph given in option(C), external pressure is isothermally increased, then last trace of liquid will disappear at point Z.

Space for Rough Work

5. Which of the following reaction(s) produce aldehyde as a product :



6. Which reaction(s) among the following is not represent correct product formation :



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
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Paragraph for Questions 7 and 8

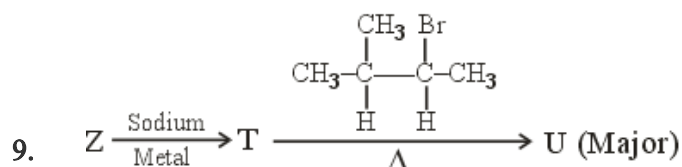
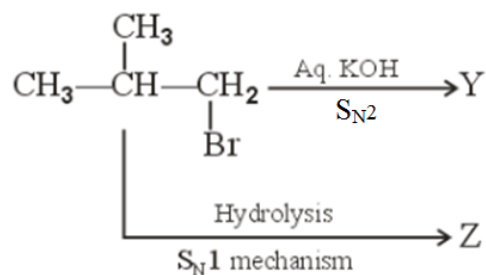
The complex HgCo_4SCN has two isomers X and Y (having Co as central metal ion in both) which are HOMOLEPTIC and magnetic moment (μ) of X = Y.

7. Choose the incorrect option for X and Y?
- (I) X and Y are different in structure around central metal ion
 (II) Oxidation of Hg and Co are same in both the complexes
 (III) X has spin only magnetic moment equals to 1.73 BM
 (IV) Both isomers having same hybridisation around central metal ion
- (A) I and II (B) III and IV
 (C) I and III (D) II, III and IV
8. The individual sum of the linkage isomers of 'X' and 'Y' are.
- (A) 8 (B) 6 (C) 10 (D) 4

Space for Rough Work

Paragraph for Questions 9 and 10

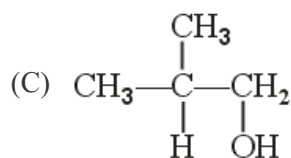
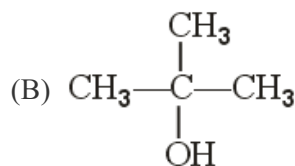
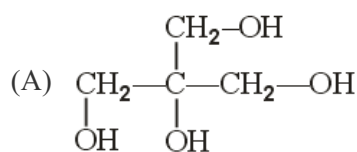
Alkyl halides can undergo SN^1 , SN^2 , E_1 , E_2 , E_{1cb} depending on various conditions.



Select the correct statement :

- (A) Formation of U follows E^2 mechanism to give saytzeff product
- (B) Formation of U follows E_{1cb} mechanism to give saytzeff product
- (C) Formation of U follows E^1 mechanism to give Hoffmann product
- (D) Formation of U follows E^2 mechanism to give Hoffmann product

10. Identify 'Y' :



(D) Both (B) & (C)

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Find number of process or ore given in Column-I are correctly represented with Column-II. You have to consider P for P, Q for Q, R for R etc.

(I) Process/Ore		(II) Related to	
(P)	Matte smelting	(P)	Cu
(Q)	Cyanide leaching	(Q)	Au
(R)	Mond's process	(R)	Reaction of metal + H ₂
(S)	Fused oxide electrolysis	(S)	Al
(T)	Chromite ore	(T)	Fe + Cr
(U)	Van Arkel process	(U)	Zr
(V)	Bacterial leaching	(V)	Low grade Cu ore
(W)	Separation of FeWO ₄	(W)	Extraction of Pb

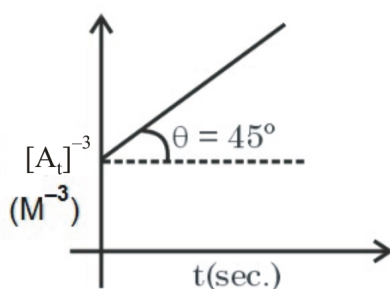
Space for Rough Work

2. Consider the following statements

- (1) $\text{CO}_2(\text{g}) + \text{C}(\text{s}) \rightarrow 2\text{CO}(\text{g})$; it is an exothermic reaction
 - (2) Ilmenite is an carbonate ore containing two metals
 - (3) In froth floatation process KCN act as an activator when galena contains impurity of ZnS
 - (4) Liquation is applicable for the purification of Sn, Pb, Zn
 - (5) $\text{Au} + \text{NaCN} + \text{O}_2 + \text{H}_2\text{O} \rightarrow \text{Na}[\text{Au}(\text{CN})_4]$ reaction occurs during leaching of gold
 - (6) Al_2O_3 can be obtained in solution from red bauxite by using concentrated NaOH
 - (7) MgO was reduced by C at very high temperature followed by a rapid cooling with the help of inert gas
- The number of correct statement(s) in the above given statements are :

3. For the reaction ,

$4\text{A} \rightarrow 3\text{B}$, the following graph is obtained experimentally, the rate of reaction (in M/sec) when $[\text{A}] = 2\text{M}$ is



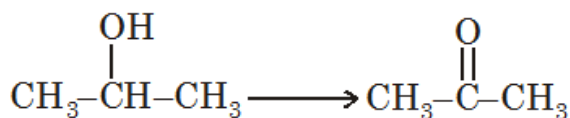
4. The decomposition of compound P, at temperature T according to the equation $2\text{P}_{(\text{g})} \rightarrow 4\text{Q}_{(\text{g})} + \text{R}_{(\text{g})} + \text{S}_{(\text{l})}$ is the first order reaction. After 30 minutes from the start of decomposition in a closed vessel, the total pressure developed is found to be 317 mm Hg and after a long period of time the total pressure observed to be 617 mm Hg. The total pressure of the vessel after 75 minute, if the volume of liquid S is supposed to be negligible is:
(Given : Vapour pressure of $\text{S}_{(\text{l})}$ at temperature T = 32.5 mm Hg, $\ln(1.169) = 0.16$, $\ln(1.5) = 0.40$)

Space for Rough Work

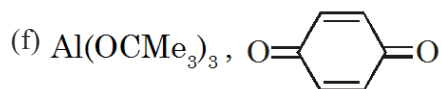
5. The saturated solution of a certain salt show elevation in boiling point equal to 3.12×10^{-3} K in aqueous solution. If the solubility of salt is 2×10^{-3} mole/kg at boiling point, then how many moles of ions are given by 1 mole of solid salt on complete dissociation.

$$(K_b(\text{H}_2\text{O}) = 0.52 \text{ K kg/mole})$$

6. If all carbonyls isomers of molecular mass = 86 u are separately treated with CH_3MgBr followed by acid hydrolysis, how many of them will give racemic mixture ?
7. How many of reagents can carry out following transformation :



- (a) Cu, 300°C
 (b) O_3 / Zn, H_2O
 (c) Zn-Hg / HCl
 (d) Jones reagent $\text{CrO}_3 + \text{H}_2\text{O}$
 (e) KMnO_4 , NaOH, Δ



- (g) Collin's reagent
 (h) (i) TsCl, (ii) DMSO, NaHCO_3 , Δ
 (i) $\text{C}_5\text{H}_5\text{HCrO}_3\text{Cl}^-$ (PCC)
8. An ideal crystal AB has rock salt structure in which A^+ is doped with atoms of 'C' which occupy all the tetrahedral voids without distorting the lattice. If the crystal shows defect such that the body-centered atoms is missing, then the percentage of body diagonal covered by ions/atoms in the defective crystal is (Given $\sqrt{1.5} = 29/24$)

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

1. Consider $P(x) = 9x^8 - 18x^5 - 20x^3 + 15$, then $P(x) = 0$ has

- (A) atleast one real root in $[0, 3^{1/3}]$
 (B) atleast one real root in $[5^{1/5}, 3^{1/3}]$
 (C) atleast one real root in $[0, 5^{1/5}]$
 (D) no real root in $[0, 3^{1/3}]$
-

Space for Rough Work

2. Let $f(x)$ be a cubic polynomial which has local maximum at $x = -1$ and $f'(x)$ has local minimum at $x = 1$. If $f(-1) = 10$ and $f(0) = 5$ and roots of $f(x)$ are α, β, γ , then $[\alpha]^2 + [\beta]^2 + [\gamma]^2$ is (where $[.]$ denotes greatest integer function)
- (A) 20 (B) 22
(C) 25 (D) 31
3. Let $f(x, y) = \frac{1}{\lim_{\alpha \rightarrow 0} \left(\frac{\sqrt[3]{\sec \alpha} - \sqrt[3]{\sec \alpha}}{\alpha^2} \right)}$, then
- (A) $f(2, 3) = 12$
(B) $f(x, y) = 12$ has infinitely many solutions
(C) Slope of tangent to the curve $f(x, y) = 12$ at $x = 2$ is 4
(D) $f(x, y) = 12$ has no horizontal tangent
4. If $y^x - x^y = 1$, then $\frac{dy}{dx}$ at $x = 1$, is -
- (A) $2(1 - \ln 2)$ (B) $2(1 + \ln 2)$
(C) $\ln(4e^2)$ (D) $\ln\left(\frac{e^2}{4}\right)$

Space for Rough Work

5. Identify the **CORRECT** statements -
- (A) Sum of two periodic functions may or maynot be a periodic function
- (B) A differentiable function in $\mathbb{R}$ will always be continuous in $\mathbb{R}$
- (C) A continuous function whose domain is in a closed interval is always bounded
- (D) In $[a, b]$ for a function $f(x)$, if $f(a) \cdot f(b) < 0$, then there exists atleast one root of $f(x) = 0$ in (a, b)
6. If $n \in \mathbb{N}$ and $f(x) = ({}^nC_0x + 1) ({}^nC_1x + 2) ({}^nC_2x + 3) ({}^nC_3x + 4) \dots ({}^nC_nx + n + 1)$ and $f'(0) = \frac{3}{2}(7!)$, then n is equal to
- (A) 3 (B) 4
- (C) 5 (D) 6
-

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- This section contains **TWO** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

Paragraph for Questions 7 and 8

Let $y = f(x)$ be a fourth degree algebraic polynomial function such that

(i) $\lim_{x \rightarrow 1} \frac{f(x)}{(x-1)^2} = 2$

(ii) It cuts y-axis a point where slope of normal is $\frac{1}{8}$

(iii) Tangent at $x = 2$ is parallel to $y = 8x + 1$

Consider $y = g(x)$ be a quadratic curve touching and cutting $y = f(x)$ on x-axis and y-axis respectively.

7. The angle of intersection between $y = f(x)$ & $y = g(x)$ at point where they cut y-axis is -

(A) $\tan^{-1} \frac{2}{33}$ (B) $\tan^{-1} \frac{2}{49}$ (C) $\tan^{-1} \frac{14}{33}$ (D) $\tan^{-1} \frac{14}{49}$

8. The abscissae of point on curve $y = g(x)$, where length of subtangent is 2018 can be -

(A) 1010 (B) 2019 (C) 3028 (D) 4037

Space for Rough Work

Paragraph for Questions 9 and 10

Let $f(x)$ be a polynomial function satisfying the following conditions $\lim_{x \rightarrow \infty} \frac{f(x)}{|x|^3} = 0$, $\lim_{x \rightarrow \infty} (\sqrt{f(x)} - x) = -1$ and $f(0) = 0$ (Note : $[k]$ denotes greatest integer function less than or equal to k)

On the basis of above information, answer the following questions :

9. The number of points of discontinuity of $g(x) = [f(x)]$ in $[0, 3]$ is

(A) 2 (B) 3

(C) 4 (D) 5

10. The number of points of non-derivability of $h(x) = |f(|x|)|$ is -

(A) 2 (B) 3

(C) 4 (D) 5

Space for Rough Work

SECTION-II : (Maximum Marks: 24)

- This section contains **EIGHT (08)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

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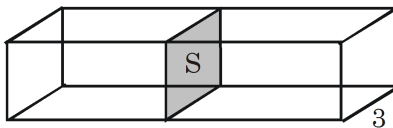
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1. The number of tangents to the curve $y = \cos(x + 2y)$, $-2\pi \leq x \leq 2\pi$, that are parallel to the line $x + 3y = 0$ is
 2. Let $f(x)$ be a differentiable function on $[0, 8]$ such that $f(1) = 5$, $f(2) = \frac{1}{2}$, $f(3) = 6$, $f(4) = -3$, $f(5) = 4$, $f(6) = \frac{1}{4}$ and $f(7) = \frac{-1}{5}$. If the minimum number of roots of the equation $f'(x) - f'(x)(f(x))^2 = 0$ is λ , then $\frac{\lambda}{3}$ equals
-

Space for Rough Work

3. Let $f(x)$ be a real valued function satisfying $2f(xy) = (f(x))^y + (f(y))^x \forall x, y \in \mathbb{R}, f(1) = 3$ and $\lim_{x \rightarrow 0} \frac{f(x) + f(-x) - f(f(0)) + 1}{x^2}$ is equal to $(\ln k)^2$ where k is natural number, then the value of $\frac{k}{2}$ is
4. $\lim_{x \rightarrow \infty} x^3 \left(\sqrt{x^2 + \sqrt{1 + x^4}} - \sqrt{2x} \right) = \frac{\sqrt{2}}{\lambda}$ find value of λ
5. Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function differentiable at 3 and satisfying $f(3) = 3f'(3)$. If the value of $\lim_{x \rightarrow \infty} \left(\frac{f\left(3 + \frac{3}{x}\right)}{f(3)} \right)^x$ is L then $\frac{[L]}{5}$ is (where $[.]$ is greatest integer function)

Space for Rough Work

6. Let $f(x) = (x-1)^{100}(x-2)^{2(99)}(x-3)^{3(98)} \dots (x-100)^{100}$. If $P = \frac{f'(101)}{f(101)}$, then $\frac{P}{5000}$ equals
7. If a rectangular tank has a sliding panel S that divides it into two adjustable tanks of width 3 ft. See figure. Water is poured into the left compartment at the rate of $5 \text{ ft}^3/\text{min}$. At the same time S is moved to the right at the rate of $3 \text{ ft}/\text{min}$. When the left compartment is 10 ft long it contains 70 ft^3 of water, then how fast the water level is falling



8. If $f(x) + f\left(1 - \frac{1}{x}\right) = 1 + x \quad \forall x \in \mathbb{R} - \{0, 1\}$, then value of $f(2)$ is

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work